

Chris Dollo

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Education

University of Maryland, Baltimore County

Catonsville, MD

Master of Science in Computer Science

Jan 2026 – Dec 2027

Bachelor of Science in Computer Science

Jan 2022 – Dec 2025

- Relevant Coursework: Advanced Algorithms, Software Engineering, Machine Learning, Natural Language Processing

Experience

Machine Learning Intern | Verizon | Catonsville, MD

May 2025 – Aug 2025

- Designed and deployed an end-to-end machine learning pipeline using YOLOv8, translating business requirements into technical specifications and achieving 94% accuracy on 14,000+ images.
- Preprocessed diverse datasets from Roboflow and Hugging Face, improving model robustness through data augmentation (rotation, scaling, brightness/contrast, saturation).
- Optimized training workflows and inference performance, reducing false detections by 20% and enhancing model reliability for real-time inspection use cases.
- Developed automated unit and integration tests, applying quality assurance and test-driven development principles to validate production workflows.

Undergraduate Research Assistant | UMBC Vinjamuri Lab | Catonsville, MD

Jan 2024 – Present

- Designed and implemented a real-time Python system for gesture analysis, translating user requirements into technical system specifications
- Extracted and modeled joint landmarks and Gaussian-based joint angular velocities from over 50 mudras and natural hand grasps to derive “syllables of motion.”
- Achieved 95.78% reconstruction accuracy for natural grasps and 92.99% for mudras, outperforming baseline models (88.92% and 82.51%, respectively).
- Developed and evaluated machine learning models for sensor-based gesture classification, performing system debugging, performance analysis, and documentation.
- **Publication:** Chris Dollo, Parthan Olikkal, Akshara Ajendla, and Ramana Vinjamuri. “Reconstructing Hand Gestures with Synergies Extracted from Dance Movements.” In Nature, Scientific Reports, 2025. (Accepted)

Projects

Hellacious | Social Media App | React.js, Flask, MongoDB, Firebase, AWS S3

Oct 2025 – Present

- Designed and implemented a full-stack social media application following the software development lifecycle (SDLC), translating user requirements into production features.
- Integrated authentication and backend services using RESTful APIs, applying system and technology integration principles.
- Deployed cloud-based services on AWS S3 and implemented logging, error handling, and performance monitoring.
- Utilized **Tailwind CSS** for responsive design and **Git** for version control and CI/CD collaboration.

EcoRecyclr | AI-Powered Waste Classification System | PyTorch, Flask

Sep 2025 - Oct 2025

- Led a 4-member team during the Booz Allen Hamilton Code for Good hackathon and won 1st place for developing an Android app that classifies waste items to determine recyclability.
- Achieved 95% accuracy using a ResNet-18 deep learning model to classify waste materials in real time.
- Implemented cloud-native APIs and evaluated new technologies for system fit and downstream performance impact.
- Utilized background services to handle model inference and image preprocessing efficiently without blocking the UI thread, ensuring smooth user interaction.

GradPath | Student Graduation Planner | React.js, Flask, PostgreSQL, Docker, AWS

Jan 2025 - May 2025

- Engineered cloud-based microservices using Flask and Docker, enabling scalable applications integration across services.
- Developed a React.js frontend and REST APIs supporting dynamic data queries with PostgreSQL.
- Followed release management processes using Git-based CI/CD pipelines and Docker Compose.

Technical Skills

Languages: C/C++, Python, Java, Kotlin, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks & Tools: React, Next.js, Tailwind CSS, Flask, Node.js, Docker, Git, AWS (EC2, S3, Lambda)

Core Concepts: Software Development Lifecycle (SDLC), System Design, Algorithms & Data Structures, Microservices, Cloud Computing, Applications Integration, Debugging & Troubleshooting, Quality Assurance & Testing, RESTful APIs